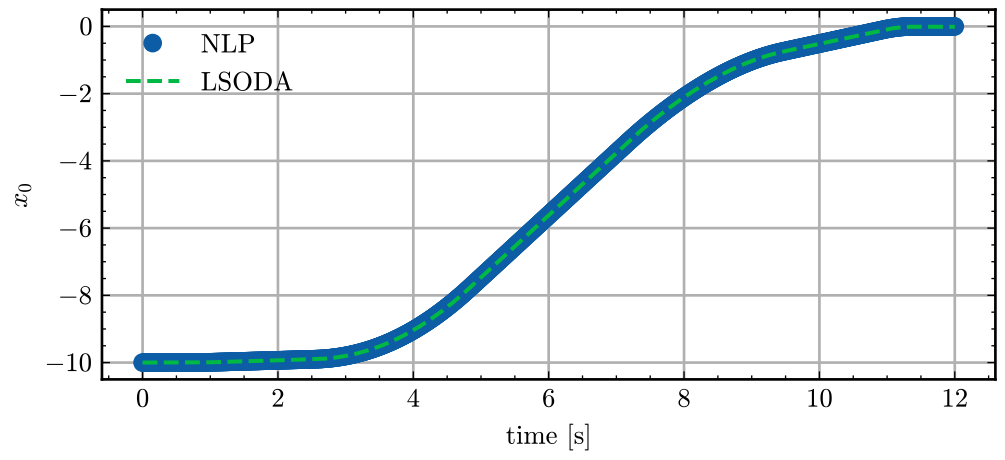
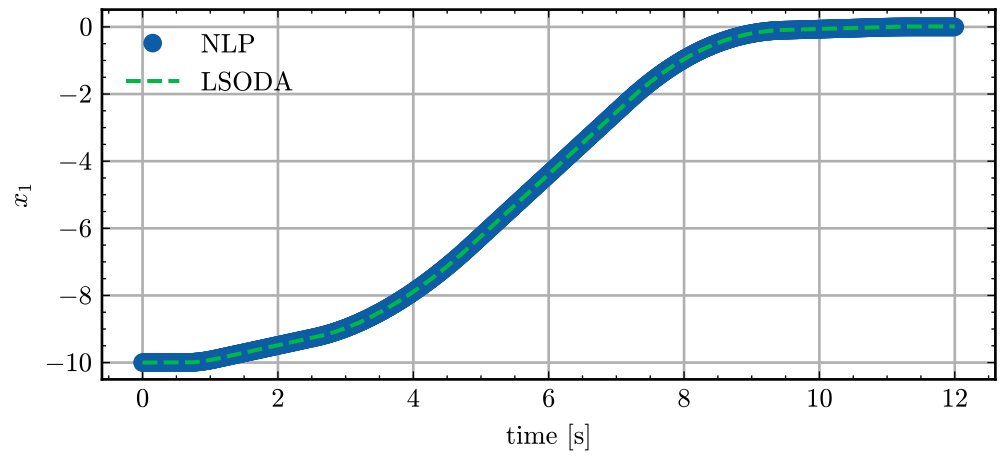
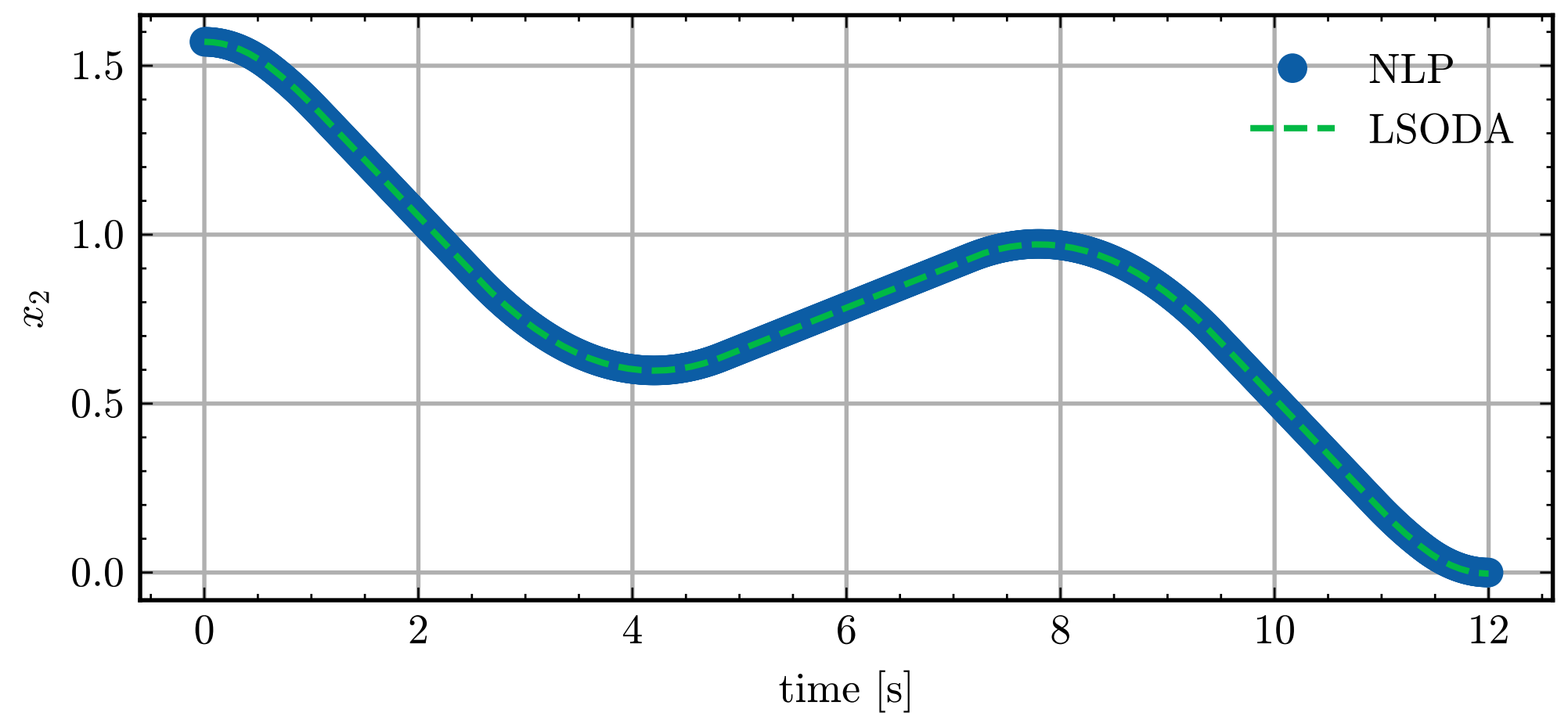
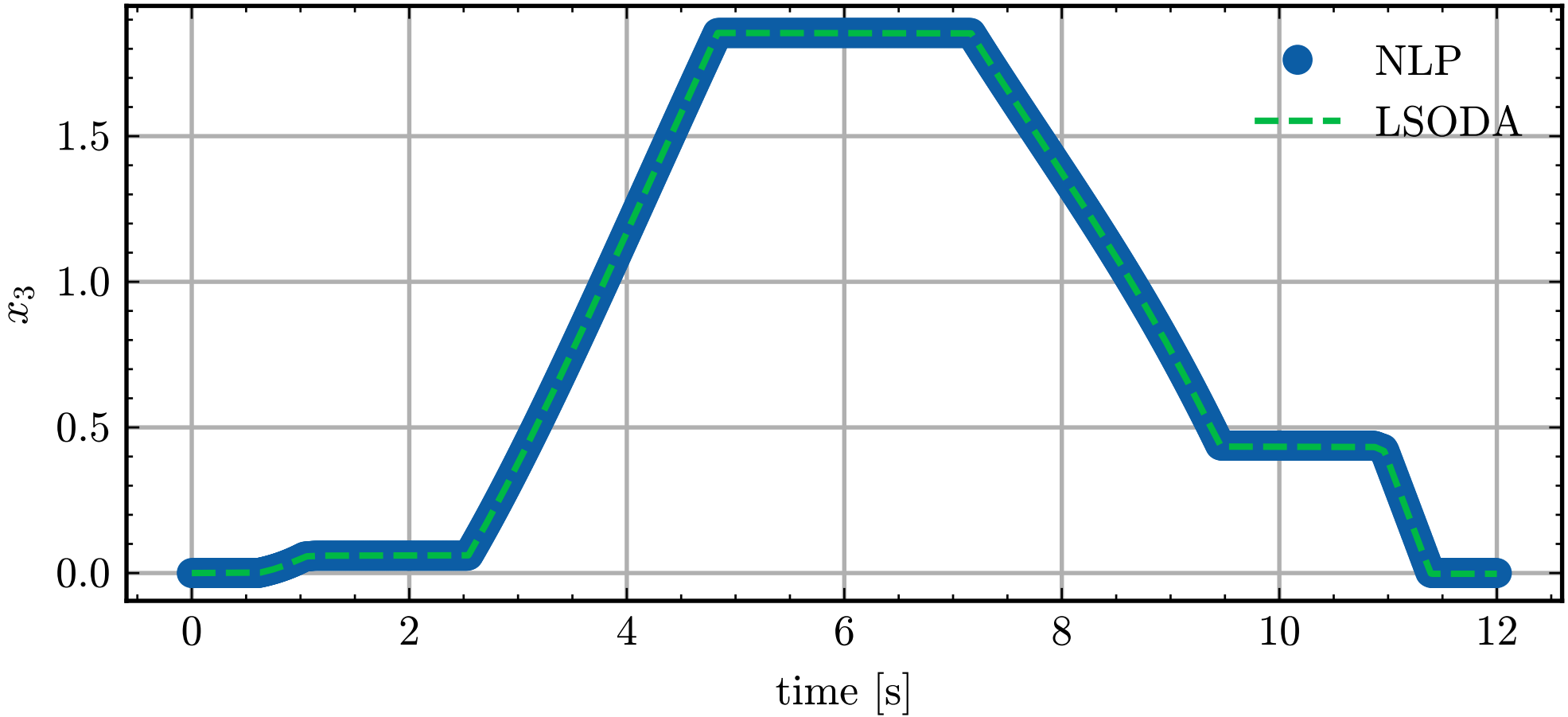
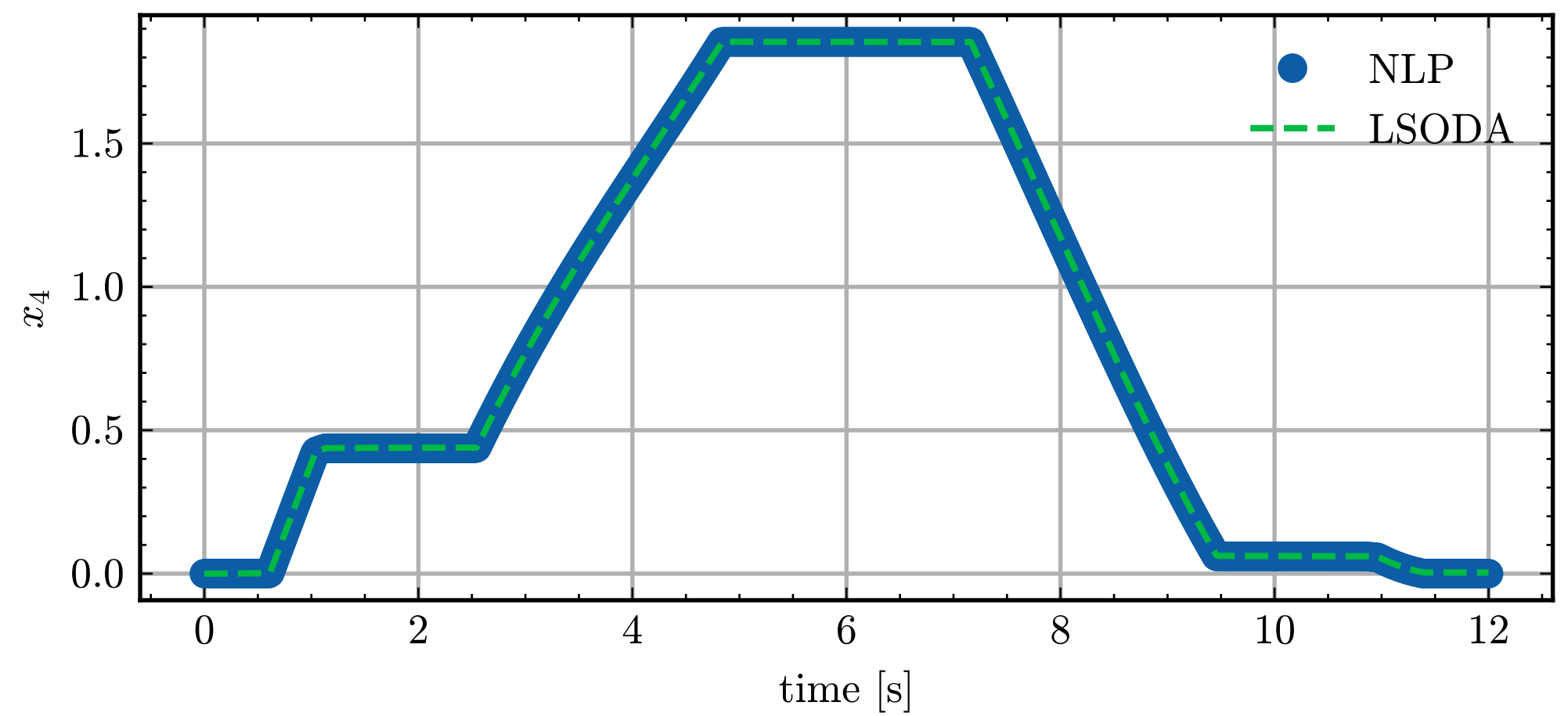
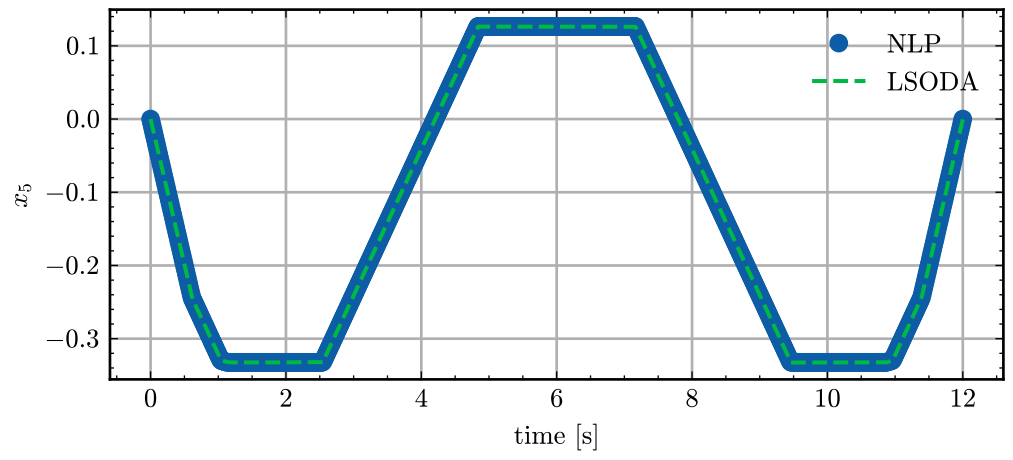

NLP
LSODA
x_0
0
−2
−4
−6
−8
−10
0
2
4
6
8
10
12
time [s]

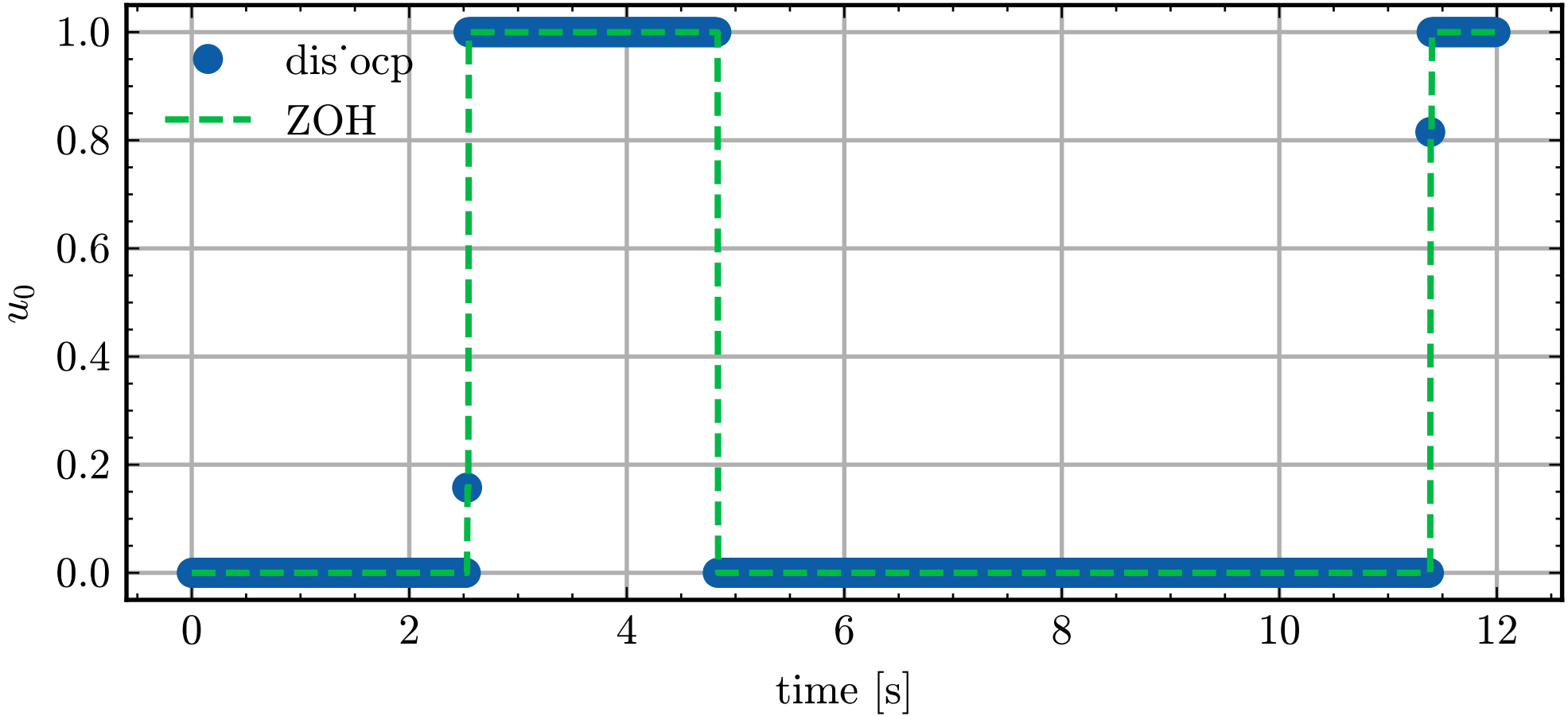

NLP
LSODA
x_1
0
−2
−4
−6
−8
−10
0
2
4
6
8
10
12
time [s]

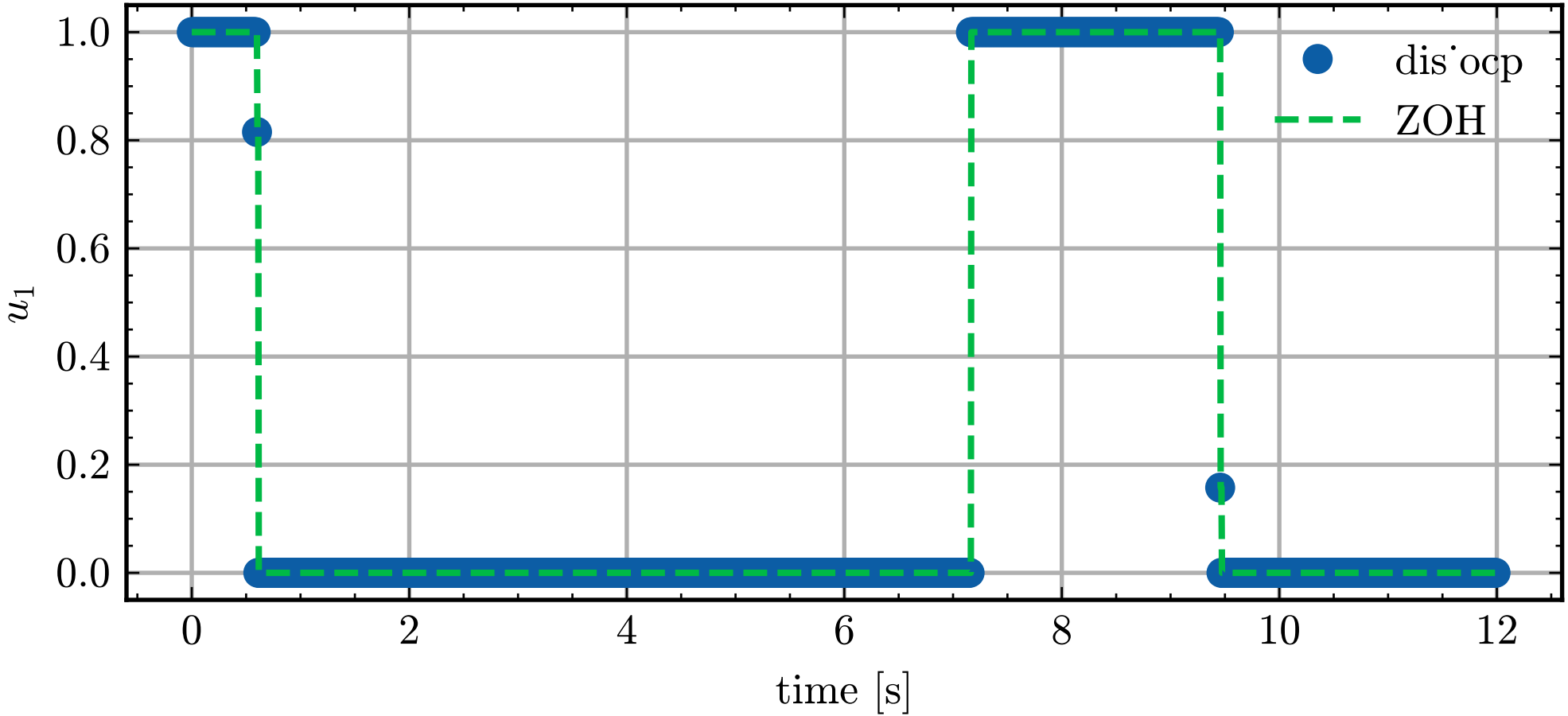

1.5
1.0
0.5
0.0
x_2
0
2
4
6
8
10
12
time [s]
NLP
LSODA

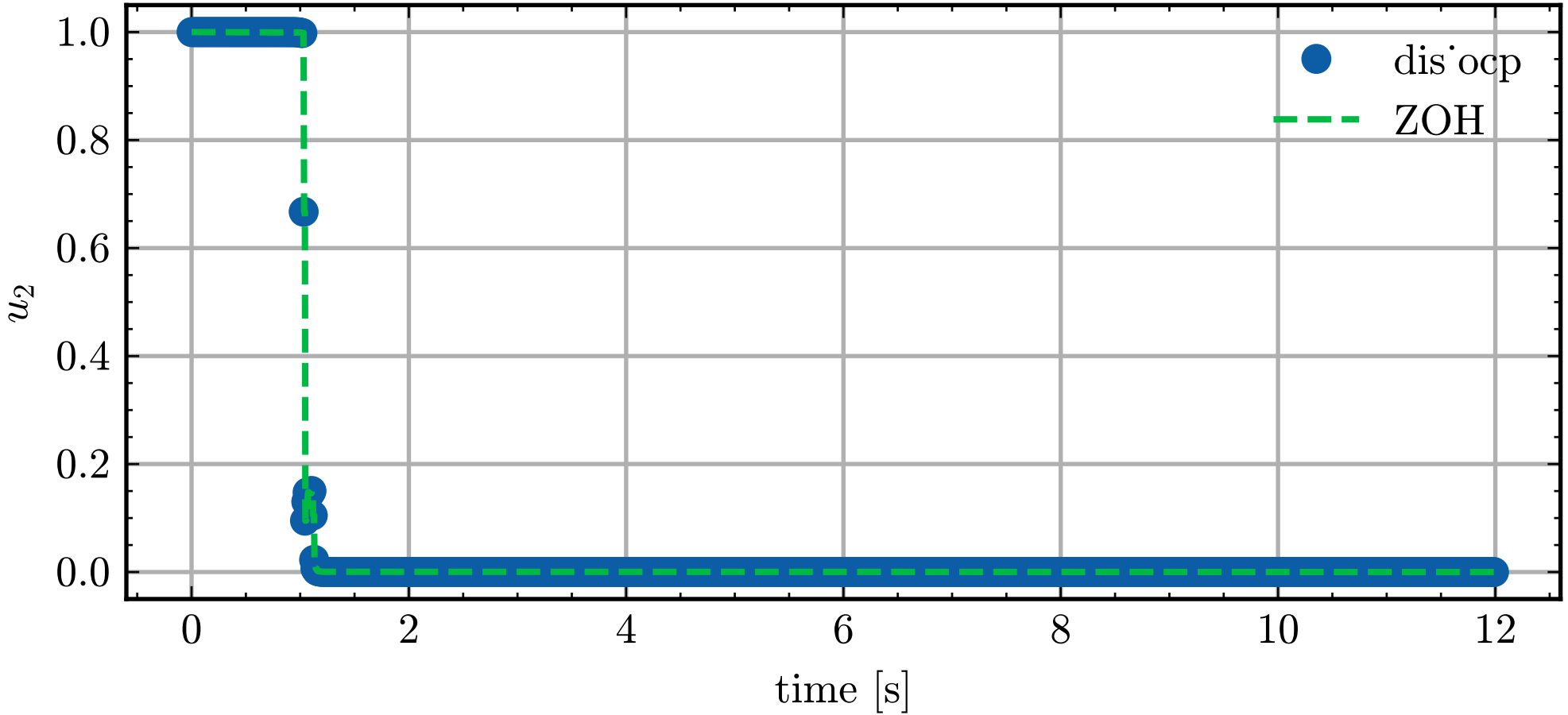

NLP
LSODA
x_3
0.0
0.5
1.0
1.5
0
2
4
6
8
10
12
time [s]

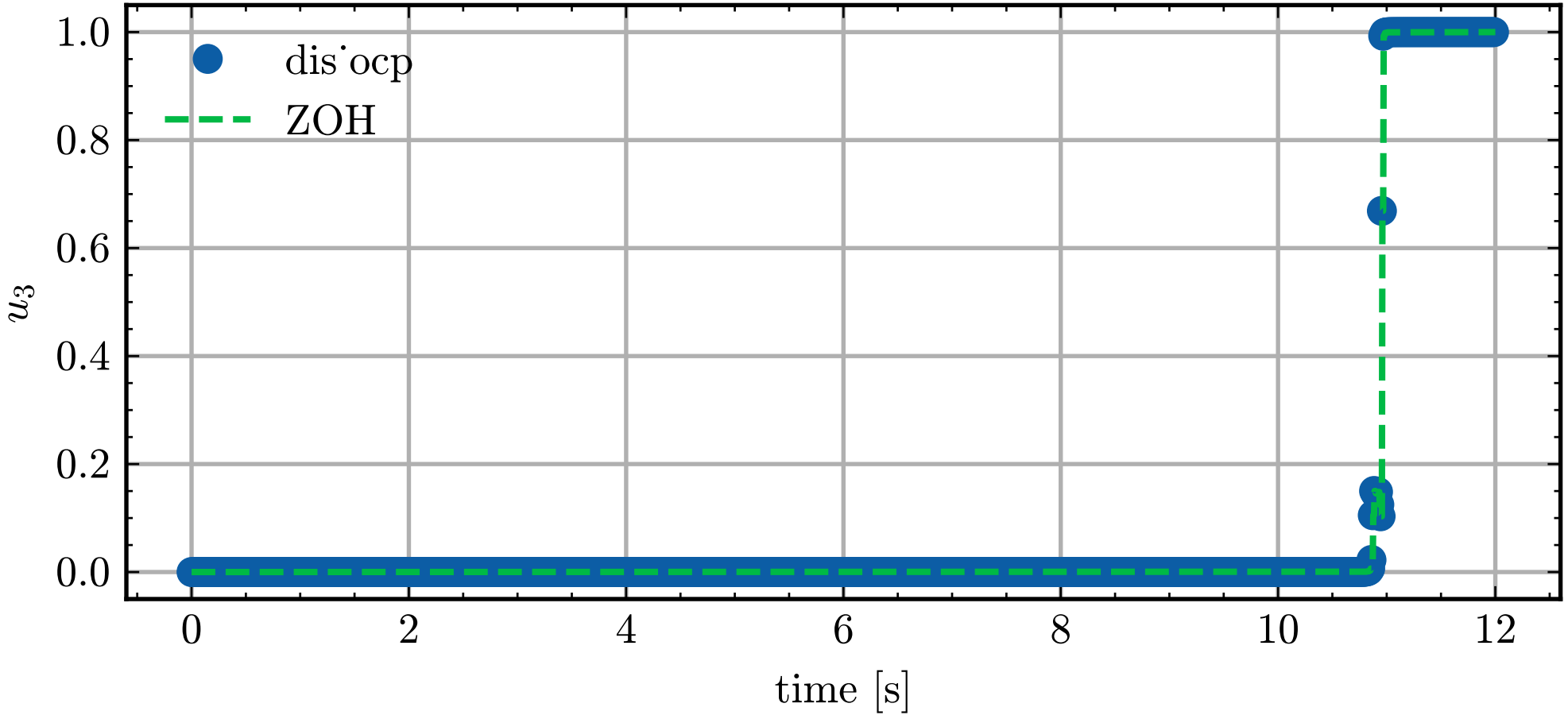

NLP
LSODA
x_4
0.0
0.5
1.0
1.5
0
2
4
6
8
10
12
time [s]


NLP
LSODA
x_5
0.1
0.0
−0.1
−0.2
−0.3
0
2
4
6
8
10
12
time [s]


dis˙ocp
ZOH
1.0
0.8
0.6
0.4
0.2
0.0
u_0
0
2
4
6
8
10
12
time [s]


1.0
0.8
0.6
0.4
0.2
0.0
u_1
0
2
4
6
8
10
12
time [s]
dis˙ocp
ZOH


1.0
0.8
0.6
0.4
0.2
0.0
u_2
0
2
4
6
8
10
12
time [s]
dis˙ocp
ZOH


dis˙ocp
ZOH
1.0
0.8
0.6
0.4
0.2
0.0
u_3
0
2
4
6
8
10
12
time [s]